

Vedant Vardhaan

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EDUCATION

University of California - San Diego (UCSD)

Bachelor of Science in Data Science and Finance | **GPA: 3.96**

La Jolla, CA

Sep 2023 – Jun 2027

EXPERIENCE

Technology Consultant Intern

Jun 2025 – Sep 2025

PricewaterhouseCoopers (PwC India)

Gurugram, India

- Designed and deployed an enterprise-grade multi-agent AI system using LangGraph, Gemini 2.5 Pro, LangChain, and MCP to automate AWS and Azure cloud onboarding for enterprise clients, cutting provisioning time by 60%.
- Engineered intent detection and dynamic workflow routing using LLM-powered NLP pipelines, enabling context-aware automation across 15+ cloud services with a 45% reduction in configuration errors.
- Architected secure, modular agent workflows for credential management, IAM provisioning, and infrastructure configuration, delivered as a production system in live PwC client engagements.

Engineering Manager & Software Engineer

Nov 2024 – Present

CSES Open-Source

La Jolla, CA

- Leading development of an AI-powered financial assistant within TritonSpend, a mobile app actively used by 100+ UCSD students, integrating personalized spending insights and goal recommendations.
- Architected the backend AI layer using Node.js and PostgreSQL, designing data pipelines that feed real-time transaction signals into the recommendation engine.

Undergraduate Researcher - Project Lead

Oct 2025 – Apr 2026

Engineers for Exploration (E4E) @ UC San Diego

La Jolla, CA

- Built and deployed a production ML pipeline fusing drone imagery and Sentinel-2 multispectral satellite data to predict mangrove presence across coastal regions, processing 10+ GB of geospatial data.
- Designed the full model architecture using TensorFlow and Google Earth Engine APIs, covering data ingestion, feature fusion, and inference, improving classification accuracy by 25% over baseline.
- Delivered the system to the Scripps Institution of Oceanography for active use in real-world coastal conservation monitoring.

ML Research Intern

Jun 2024 – Sep 2024

Indian Institute of Technology (IIT) Guwahati

Guwahati, India

- Engineered a ResNet50-based CNN for SAR satellite image classification across 4 land cover categories, achieving 80% accuracy through iterative architecture tuning and data augmentation.
- Applied Lee and Gamma-MAP speckle filters to denoise SAR imagery and improve feature clarity, evaluating model robustness with precision, recall, F1-score, and confusion matrices.

PROJECTS

InvasiveWatch: Global Species Tracker | Python, FastAPI, FAISS, Scikit-learn, Mapbox

Jan 2026 – Mar 2026

- Shipped a full-stack AI platform scoring invasive species risk for any location, pairing a hybrid ML engine with FAISS vector search over 96K+ species (83.4% recall) across climate, aggression, and invasion-history signals.
- Integrated live biodiversity APIs into the inference pipeline for real-time species filtering, awarded **Best Project, DS3 Winter 2026** out of 10 competing teams.

SoCal Guessr | Python, PyTorch, ResNet18, Transfer Learning

Feb 2026 – Mar 2026

- Fine-tuned an ImageNet-pretrained ResNet18 with a custom 6-class classification head to predict which Southern California city a street photo was taken in, using transfer learning on labeled location images.
- Built an end-to-end training, checkpointing, and inference pipeline with visual diagnostics for reliable city-level predictions.

MarketScope: Intelligent Stock Forecasting | Python, LSTM/GRU, TensorFlow, Streamlit

Oct 2024 – Jan 2025

- Deployed a hybrid LSTM-GRU forecasting system with Monte Carlo simulations for probabilistic stock price prediction, integrating VADER sentiment analysis on financial news and risk metrics (VaR, Sharpe ratio).
- Shipped as a modular Streamlit app with real-time yFinance ingestion and multi-modal signal fusion, covering the full pipeline from raw data to deployed inference.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, R

AI & ML: TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, OpenCV, FAISS, RAG, Reinforcement Learning, Cursor, Claude

GenAI & Agents: LangChain, LangGraph, Gemini, MCP, Prompt Engineering, Multi-Agent Systems

MLOps & Deployment: Streamlit, FastAPI, REST APIs, Docker, CI/CD, Git/GitHub, Google Earth Engine

Cloud & Infra: AWS (EC2, S3, IAM), Azure, GCP, PostgreSQL, Node.js, Supabase

Data: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Apache Spark